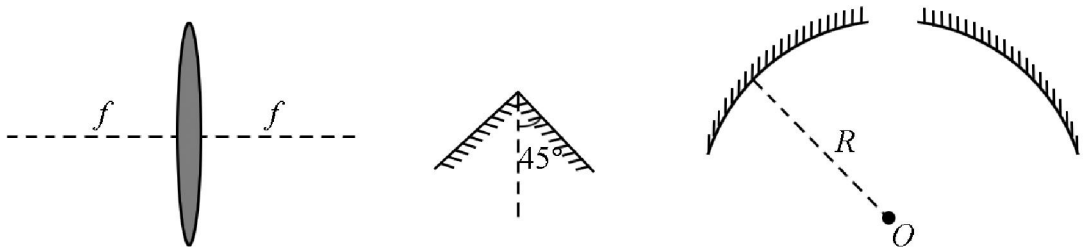
VIII. (10 points) A hollow laser beam is cylindrical with zero intensity on its axis. It traps tiny particles at the center, acting as optical guides, tweezers or spanners for precise control of nanoparticles and biological cells. Widely used in biology, laser processing and atomic cooling, it is becoming a new subdiscipline.

Optical systems turn solid cylindrical beams into hollow beams. Use a convex lens of focal length $f$, a mirror with cone angle 45°, and a spherical mirror of radius $R$ with a central circular hole, as shown. Use them to design an optical system converting a very thin solid cylindrical input beam into a hollow output beam with a cylindrical empty core of radius $r$. Answer:

1. Draw the ray diagram.
2. Find distance $x$ from the cone apex to the spherical mirror's center of curvature.




(a) (b) (c)